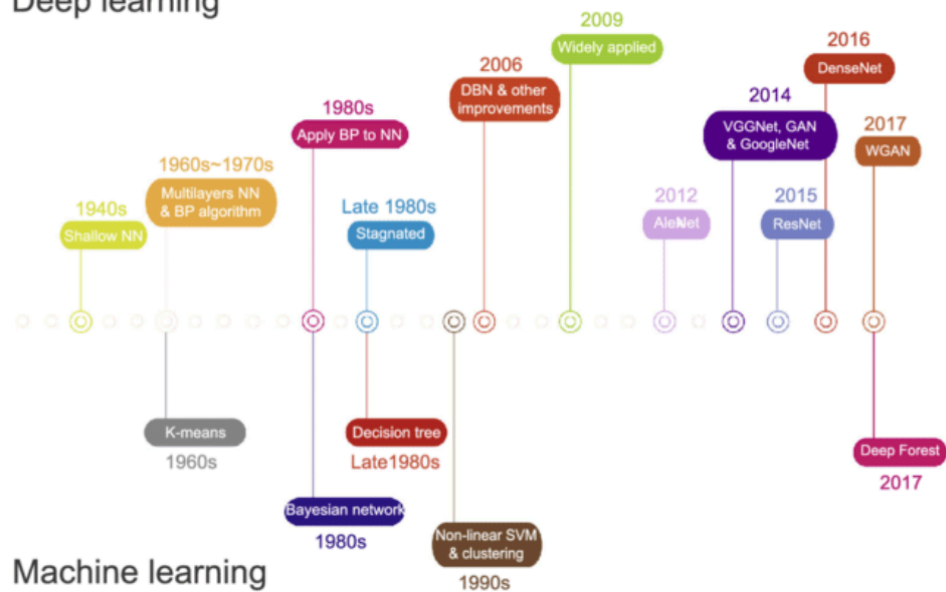


Select one modern deep learning technique, model, or architecture not included in the provided historical timeline. Briefly explain its purpose, key innovation, how it works, and at least one real-world application. Support your discussion with credible references.

Deep learning



Swin Transformer

A Modern Deep Learning Architecture

One modern deep learning architecture that is not included in the given historical timeline is the Swin Transformer, introduced by Liu et al. in 2021. It was developed mainly for computer vision tasks and addresses some of the limitations of applying the original Transformer architecture directly to high-resolution images. Its purpose is to learn visual features efficiently while still capturing relationships between different parts of an image. Unlike traditional convolutional neural networks, Swin Transformer uses a Transformer-based attention mechanism to understand visual information at different scales (Liu et al., 2021).

The main innovation of Swin Transformer is its shifted-window self-attention. Instead of calculating self-attention across every image patch at the same time, the image is divided into smaller non-overlapping windows, and self-attention is performed within each window. In the following layer, the windows are shifted so that information from neighboring windows can interact. This reduces the computational cost while still allowing the model to learn relationships across different regions of an image. The architecture also gradually merges image patches as the network becomes deeper, creating a hierarchical representation that can work with objects and features at different scales.

Swin Transformer can be used for several real-world computer vision applications, including image classification, object detection, and semantic segmentation. The original study demonstrated its effectiveness on datasets such as ImageNet, COCO, and ADE20K. Its official implementation also supports applications such as instance segmentation and video action recognition. Because of its ability to process visual information efficiently at different scales, Swin Transformer has become a useful architecture for modern computer vision systems.

Reference

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., & Guo, B. (2021). *Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows*. Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 10012–10022.
<https://doi.org/10.1109/ICCV48922.2021.00986> (The Computer Vision Foundation)